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Big Data**A Survey Research Perspective***Reg Baker**Marketing Research Institute International, Ann Arbor, MI, USA***3.1 Introduction**

In 1987, to commemorate its 50th anniversary, *Public Opinion Quarterly* asked 16 well-known scholars and survey practitioners to offer their visions of the future of public opinion research (Bogart, 1987). In his response, Harold Mendelsohn described how “the eventual ‘computerizing’ of the American home undoubtedly will contribute significantly to the speed, accuracy, and economy with which data will be gathered, analyzed, and readied for dissemination.” James Beniger wrote that “a host of new technologies will ... make possible the real-time mass monitoring of individual behavior ... Survey research will increasingly give way to more direct measures of behavior made possible by new computer-based technologies.” In a particularly chilling vision, and perhaps somewhat tongue-in-cheek, Robert Worcester described:

Market researchers are close to their tactical ideal, the comprehensively wired micro model of segmented household “norms” which can be conceptualized, pressurized, test marketed to, weighted (up and down), copy tested, product tested, studied, and, yes, manipulated by cables, satellites, and sensors (worn in rings, necklaces, earrings, or even—shades of George Orwell—implanted!).

Three decades later this future, if not now, is at least more clearly visible. This is the world of big data, and depending on where you sit and the type of research you do, it is either a dream come true or the apocalypse.

This chapter explores the big data phenomenon in five sections. The first two describe aspects of big data that are especially challenging for survey researchers—the size and complexity of the databases and the analytic approaches favored by data scientists. This is followed by a discussion of the challenges of assessing data quality when working with big data and then an overview of applications in market, opinion, and social research. The chapter concludes with a brief mention of two ethical considerations when using big data and some musings about the future of surveys in a big data world.

3.2 Definitions

There are many ways to define big data. Press (2014), for example, counted 12 different definitions. The Berkeley School of Information asked 40 thought leaders across a wide variety of industries (including publishing, fashion, food, automobiles, medicine, and marketing) how they would define the term. Not surprisingly, they got more or less 40 different definitions (Dutcher, 2014). The word cloud in Figure 3.1 summarizes those definitions.

Ward and Barker (2013) attempted to resolve some of the ambiguity in definitions of big data by reviewing those most often used by various players in the big data ecosystem of IT consultants, hardware manufacturers, software developers, and service providers. They noted that most definitions touch on three primary attributes: size, complexity, and tools. They also suggested a definition that, while not especially elegant, may serve notice to those about to enter the world of big data: “Big data is a term describing the storage and analysis of large and or complex data sets using a series of techniques including, but not limited to: NoSQL,¹ MapReduce,² and machine learning.” We might simplify that to say, *big data is a term describing the storage and analysis of datasets so large and complex that they cannot be processed or analyzed with conventional software systems.*

There are other, more intuitive definitions. In his visionary book, *Being Digital*, MIT's Nicolas Negroponte (1995) described a future in which most if not all of our interactions with other people, institutions, and even things are recorded and stored digitally where they can be retrieved, matched, and manipulated. He used the word "digitization" to describe the process by which phenomenon of all kinds would be stored as "bits rather than atoms." More recently



Figure 3.1 Word cloud based on Dutcher (2014).

1 A database design technique that overcomes some of the processing inefficiencies of classic relational databases.

2 A programming model for distributed processing of large datasets across a cluster of computers to reduce redundancies and aggregate data to a higher level.

some have used the term “datafication” to mean essentially the same thing (Mayer-Schonberger and Cukier, 2013). Writing in *The Financial Times*, British economist Tim Harford (2014) used the term “found data,” described as “the digital exhaust of web searches, credit card payments, and mobiles pinging the nearest phone mast.” Groves (2011) coined the term “organic data” and described it as the data produced by “systems that automatically track transactions of all sorts.”

3.2.1 Sources

As the forgoing suggests, big data is ubiquitous and almost any dataset a researcher might find useful has the potential to be part of big data. While there are many potential sources, we can distill them down into three broad categories.

- 1) *Transaction data* describe some event, such as when a person interacts with a business or government entity. Customer data and government administrative data are two common examples.
- 2) Social media data are drawn (or scraped) from social media networks, blogs, microblogs, ecommerce reviews, web searches, and other sources of free-form text and nontraditional data such as photographs or video recordings. Google Flu Trends is one oft-cited example (Butler, 2013). Another is the work of researchers at Carnegie Mellon University demonstrating how some relatively simple sentiment analysis of Twitter posts could be used to measure consumer confidence and presidential approval (O'Connor et al., 2009). Also see Daas and Puts (2014) and Chapter 2 for further discussion of research using Twitter.
- 3) The Internet of Things (IoT) is the least developed source. The term was coined by MIT scientist Kevin Ashton (Wood, 2015) and refers to data collected from interconnected devices covering a wide range of objects including autos, household appliances, scanners, traffic lights, security cameras, wearable sensors, GPS locators, and so forth. For example, cities around the world are using networks of roadside scanners, Bluetooth sensors, toll card tags, and signals from GPS devices in cars, trucks, and busses, along with historical data on traffic patterns and even crowdsourced incident reports from drivers to manage traffic flows, reduce congestion, and shorten driving times (see, e.g., Bertolucci, 2013; Mullich, 2013; University of Southern California Viterbi School of Engineering, 2013). These same data can be used by national statistical organizations to gather data on the movement of people and things (Daas et al., 2013). In another example, energy utility companies have been installing smart meters in U.S. homes to measure and report on electricity use enabling those utilities to understand patterns at different times of day, with changing weather, and how those vary by neighborhood and demographics (PG&E, 2015). As smart appliances such as washing machines, refrigerators, coffee makers, alarm clocks, and so forth come to market, they will generate data from which analysts can infer new information not just about electricity use but about lifestyles and the rhythms of daily life. IoT is the least deployed but potentially most far reaching of the three sources of big data.

3.2.2 Attributes

Many discussions of big data emphasize its attributes, expressed in some number of Vs. META Group (later Gartner) consultant Doug Laney started this ball rolling in 2001 when he noted that changes in the volume and character of data required a rethinking of how to manage and extract value from them (Laney, 2001). He described three characteristics of what he called, “the data deluge:” volume, variety, and velocity. Although he never used the term, “big data,” his report generally is recognized as the beginning of the big data revolution.

A decade later the conversation about big data began to include two additional Vs: veracity (sometimes called verification) and value. No one is quite sure who introduced these new terms, although database consultant Dave Beulke (2011) is often cited. More recently, van Rijmenam (2014) added two more Vs, variability and visualization, bringing the total to seven. Let us look at each.

3.2.2.1 Volume

The big data world is one of petabytes, exabytes, zettabytes, and yottabytes requiring “massively parallel software running on tens, hundreds, or even thousands of servers” (Hopkins, 2011). Survey researchers accustomed to thinking of data in terms of megabytes or perhaps gigabytes may find it difficult to imagine just how massive big data can be and the difficulty of processing it.

A 2014 study by IDC (2014) estimated that 4.4 zettabytes of data were created or copied in 2013. The study projected that this amount would increase by 40% each year and by 2020 would constitute 44 zettabytes of data, an amount equal to 44 trillion gigabytes.

As an example, consider the evolution of point of sale (POS) systems in retail stores. Technologies that capture data on transactions within a store via scanning at checkout were first introduced in the 1970s. These early systems stored their data within the store and were used primarily to track inventory. Over time as stores became larger, the number of products carried increased, technology improved, and large retail chains became more common. Data were uploaded and pooled across stores so that the information could be used to optimize carrying costs (Revel Systems, 2014). In the 1990s retail chains began introducing loyalty cards that allowed them to attach characteristics of individual shoppers to purchases, making it possible to target offers at groups of shoppers with shared consuming habits. Today, with over 10,000 stores worldwide, Walmart captures more than a million customer transactions each hour, all of which are uploaded to a database in excess of three petabytes (SAS, 2012), over three million gigabytes of data.

3.2.2.2 Variety

Variety refers to the structure of the data itself. A significant amount of big data comes to us either with a very loose structure or no structure at all. A simple and familiar example of the former is a document such as a questionnaire that has been tagged to identify different questionnaire elements (e.g., question text, answer text, or skip logic). The greater share of big data is likely to be unstructured, meaning “information that either does not have a predefined data model and/or is not organized in a predefined manner” (Nemschoff, 2014). An obvious example of unstructured data is social media data (blog posts, tweets, comments, photos, videos, etc.).

3.2.2.3 Velocity

The speed at which new data are generated every minute of every day is almost unimaginable. On Black Friday (the day following Thanksgiving Day in the U.S.A. and regarded as the beginning of the Christmas shopping season) in 2012, Amazon customers ordered over 26 million items, storing each in a database with links to all other purchases made by the same customer (Cheredar, 2012). That amounted to 306 items/second. The numbers from social media sites are even more mind numbing. In 2014, the DOMO software company created the infographic in Figure 3.2 showing how much data is created every minute of every day (James, 2014). As its title says, “Data Never Sleeps.”

3.2.2.4 Veracity

All of these data are useless if they do not satisfy at least some basic requirements of accuracy. This is perhaps the greatest challenge inherent in big data. Shackett (2014) was speaking for

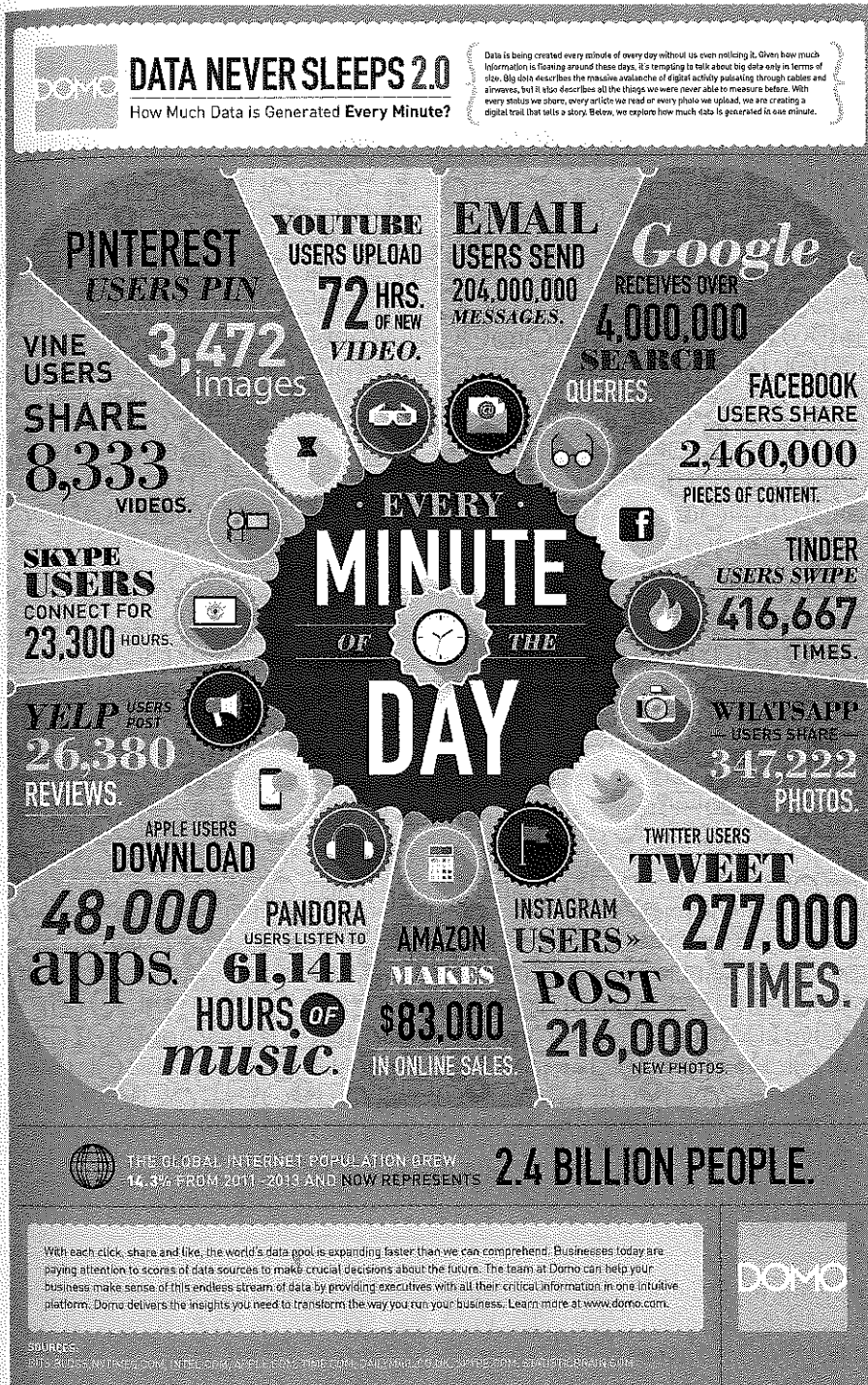


Figure 3.2 Infographic from James (2014). Reproduced with permission from Domo, Inc.

many when she described data quality as “the ugly duckling of big data.” Some of the issues that big data users confront are described in more detail later in this chapter and in Chapter 2.

3.2.2.5 Variability

Variability refers to differences in meaning across data sources (Hopkins, 2011). A significant element of big data involves processing of raw free-form text to extract meaning (often referred to as *sentiment*), and meaning can vary depending on context, tone, and a variety of other characteristics (Lazarus, 2012). Text processing systems are an essential component of big data, but there is considerable debate around how well these systems do their jobs. Grimes (2010, 2012) notes that while estimates of coding accuracy vary substantially, from as low as 30% to upwards of 70%, there is no widely accepted method by which all systems can be judged. Wilson et al. (2005) put the accuracy of human coding at just 80%, which may or may not represent a ceiling to which automated systems can aspire.

Similar challenges exist for other types of unstructured data such as photographs and video. With the exception of facial recognition and coding, developments generally lag behind those for text processing and sentiment analysis.

3.2.2.6 Value

A single large dataset (such as POS data collected in retail stores) has value in applications such as managing inventory, analyzing the impact of promotions, or describing which brands do well within a category and which do not. But the value of those data can increase exponentially, for example, when linked with loyalty card data that contain information about individual customers and purchase histories or with social media sound bites about the store experience or products purchased there. This ability to combine sources of all kinds into rich and detailed profiles of people, their behavior, opinions, and preferences, is a substantial part of the lure of big data for market, opinion, and social researchers alike.

3.2.2.7 Visualization

Finally, with large and complex datasets comes the challenge of communicating with users. Dashboards are one frequently used technique. They typically show the most important measures at a comprehensible level of aggregation, allow drill downs and custom reporting, and are continually updated as new data are added to the database.

One familiar example to many is the multilayered dashboards now routinely used by TV networks to monitor election returns. Figures 3.2 and 3.3 are good examples. Elaborate visualizations with graphics, text, and data have become increasingly popular. There is even an annual competition—the Kantar Information is Beautiful Awards—that recognizes the best visualizations each year.

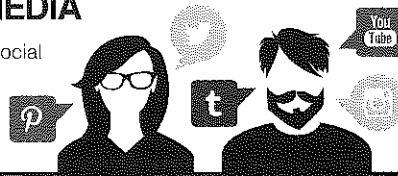
3.2.3 The Making of Big Data

As just noted, realizing the value of big data generally means linking a variety of sources together to form a combined database, often called a data warehouse. See Figure 3.4 for a somewhat idealized and simplified view of what can be a very complex, technologically intensive, and challenging process. Even in large corporations with the financial and technical means to implement big data projects, progress has been slow and sometimes disappointing (Court, 2015). Outside of the business context, research described as “big data” often uses more traditional and less technically complex methods to create datasets for analysis. Still, the general shape of activities and the risks to data quality are essentially the same.

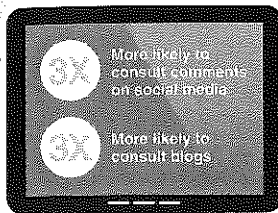
MILLENNIALS AND SOCIAL MEDIA

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Millennial shoppers trust social media more but diversify beyond Facebook



When researching products and services, Millennials are more likely than Baby Boomers to be **influenced by comments on social media or blogs**.



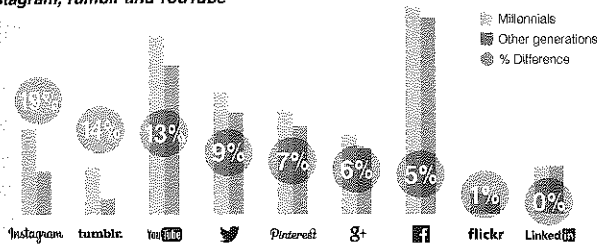
Friends and family don't have an agenda and aren't trying to sell something to me.

Blogs are usually objective, they get no financial reward and they are more knowledgeable than my friends and family.

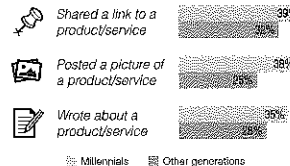
Understanding exactly how and where Millennials are doing their research has enormous implications for brand marketers.

Where are the largest differences in use between Millennials and other generations?

Instagram, Tumblr and YouTube



Millennials are **more likely to have shared a link to, posted a picture of or written about a product or service** on social media.



Millennials are more likely to **diversify beyond Facebook** and use more specialized social media platforms compared to other generations.



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Figure 3.3 Sample data visualization. Reproduced with permission from Market Strategies International.

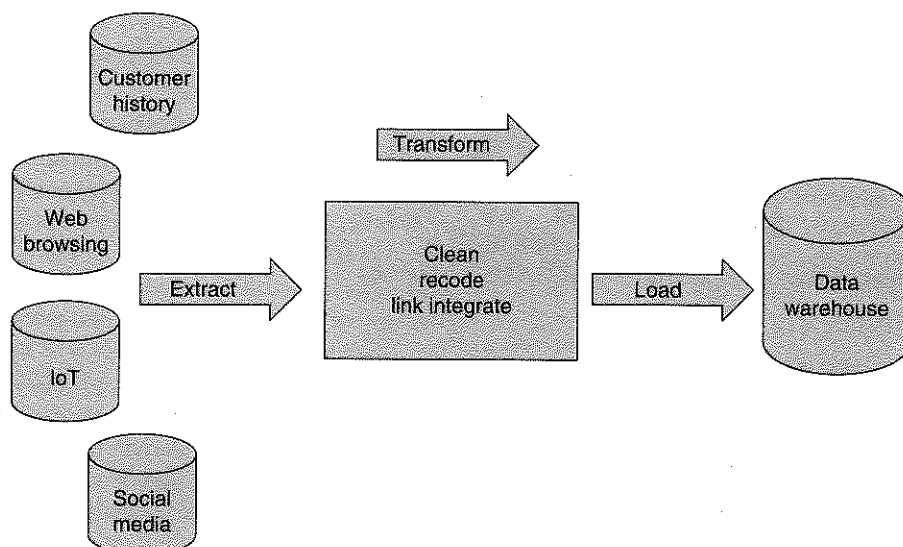


Figure 3.4 Simplified view of data linkage.

The first step is to identify and gain access to the data sources to be used. Almost by definition big data is multisourced with each of those sources having its own complexities and idiosyncrasies. At this stage, even getting access can be problematic due to issues of ownership, purpose, and confidentiality.

Even with access, data quality issues can doom the research from the outset. If we are to have confidence in big data, we need a clear view of the quality of the original data sources—the collection processes, the specific origin and meaning of individual data items (metadata³), the population covered, etc. More often than not, big data relies on data collected for some purpose other than research. The rules under which the data were collected, while adequate for their original purpose, may be inadequate for the analytic purpose or may not be complete enough to develop the rules needed to merge with other data. Transaction data may have missing data elements or not fully cover the population they were meant to cover. The variety of forms and structure of social media data, such as text, images, and video, means that they are subjected to extensive processing prior to use, opening the door to a wide range of potential errors and lost or mischaracterized meaning. IoT data are subject to a wide variety of errors in data collection due to sensor malfunction, outage, or gaps in coverage. These and other related issues are discussed in more detail in Chapter 2.

Two interrelated concepts or processes are critical in developing a full understanding of the data sources that comprise a specific instance of big data. The first is *data curation* meaning “the active and on-going management of data through its lifecycle of interest and usefulness to scholarship, science, and education” (Court, 2015). On the one hand, data curation is much like data archiving in that it preserves data over time, making reuse and analysis possible. It also can be more challenging because so many of the sources of big data were not collected for the purpose of research and therefore may not be fully documented or preserved in a format that facilitates their reuse. The data curator decides which data to preserve, assembles the necessary documentation, and ensures that the data are as described, structured for analysis, and have a mechanism

3 A description of how, when, why, by whom, and with what level of control a data item was collected.

for sharing. Given the size and complexity of many contemporary data sources, data curation often focuses on smaller datasets that can be used as inputs to merging and linking processes that create more complex databases.

The second concept is *data provenance*, meaning “the process of tracing and recording the origins of data and its movement between databases” (Buneman et al., 2000). It is especially critical when using a database constructed from multiple sources, where a number of merging, linking, transforming, or aggregating steps already have been performed. Data provenance essentially involves the reconstruction of the history of each data element in as much detail as possible.

Using data that have not been fully curated or for which full provenance is unknown can have major consequences for the research. Under these circumstances, the hard work of investigating, cleaning, and documenting data falls to the project team and can consume a major portion of the project’s resources. Even then, the full history of the data may be impossible to uncover, weaknesses in the data not fully understood, and the reliability of the results compromised.

The second step in the making of big data is the point at which datasets are merged using a series of ETLs (for extract, transform, and load), computer code that connects to some number of data sources, extracts the data items of interest, and cleans, standardizes, and loads them to a target for analysis (El-Sappaghi et al., 2011). It is the backbone of big data processing and the step at which data quality may be most at risk.

An essential input to any ETL is comprehensive metadata defining both the data items to be extracted from the different sources and the data items loaded to the target. As with any analysis, clear documentation about how a data item was defined and generated is key.

The specific operations performed by an ETL will vary depending on the sources being used and the structure and content of the target database. Those operations can include:

- Cleaning the incoming data
- Reformatting and recoding items to conform to the target’s metadata
- Coding raw data such as text
- Matching records
- Aggregating data to create roll-ups
- Standardizing missing data
- Deduplicating
- Deriving new data items

See Chapter 25 for a discussion of the difficulty and potential errors that can be introduced when linking data from different sources.

The ETL step is the point at which computing resource requirements are most intense. It relies heavily on technically complex parallel or distributed processing, a computer operation that increases the amount of computing power available to an application by splitting it into discrete parts that are run simultaneously on different processors in a cluster of computers, thereby reducing the amount of time required for computational intensive applications or those using very large databases. Apache Hadoop is a commonly used software library that supports distributed processing of these very large data sets across clusters of thousands or even tens of thousands of computers, typically in a cloud computing environment. The technical complexity of ETL processing should not be underestimated.

The foregoing discussion only scratches the surface of the complexity and difficulty of creating and maintaining ETLs. In a corporate setting where data sources may change infrequently, the level of effort invested in ETL development can be justified by their relatively long life and

continued use. In the context of a research project where the goal might be to construct a relatively static database for analysis, ETL development may consume a significant portion of the project's resources.

3.3 The Analytic Challenge: From Database Marketing to Big Data and Data Science

One of the driving interests in big data has been the potential to uncover patterns in data that can be used to predict behavior. The increasing amount of data available and the computing power to analyze it have led to the development of a set of techniques that rely on computer algorithms rather than the classic approach of collecting data to test hypothetical models. These newer techniques may be grouped under the general heading of analytics with a variety of prefixes such as "predictive," "text," "social," or "web." Despite differences among the various analytic techniques used, they typically rely on some form of machine learning.

These techniques have a considerable history. It stretches back to the emergence of database marketing in the early 1980s (Shaw and Stone, 1988) followed by the development a decade later of data mining, or as it was and still is sometimes referred to in academic circles, knowledge discovery in databases (KDD or simply KD). The essence of KD is "the systemization of methods and techniques for making sense of data" (Fayyad et al., 1996). It relies on analytic techniques based in machine learning, that is, the use of algorithms capable of finding patterns in data unguided by a set of analytic assumptions about the relationship among the data items. As Dhar and Chou (2001) describe it, "patterns emerge before reasons for them become apparent." Data miners deployed a wide range of analytic and predictive techniques including decision trees, logistic regression, neural networks, a variety of clustering techniques, and even text mining. They also have adapted these techniques to new fields including fraud detection, investment portfolio management, pattern detection in product failures, and more (Baker, 2013).

The emergence of data mining is especially significant in the context of big data because it ushered in a perspective on data analysis that is substantially different from the classic statistical analysis typically used in survey research, a perspective that today we call data science. To paraphrase Dhar (2013), classic statistical analysis asks the question: Do these data fit this model? Data mining (and by extension data science) relies on machine learning and asks a different question: What model fits these data?

This perspective has some history in academia as well. Writing in the early 1960s, Tukey (1962) suggested the use of new computer-based methods to develop hypotheses as well as test them. In 1977, the International Statistical Institute established a new section "to link traditional statistical methodology, modern computer technology, and the knowledge of domain experts in order to convert data into information and knowledge" (Press, 2013). In 1997, C. F. Jeff Wu, then chair of the Statistics Department at the University of Michigan, called for statistics to be renamed data science, statisticians to be renamed data scientists, and the curriculum in statistics to be broadened to include an emphasis on large, complex datasets and integration of other disciplines including the physical sciences, engineering, and cognitive science (Wu, 1997). In 2001, Breiman described "two cultures in the use of statistical modeling to reach conclusions from data."

One assumes that the data are generated by a given stochastic data model. The other uses algorithmic models and treats the data mechanism as unknown. The statistical community has been committed to the almost exclusive use of data models. This commitment

has led to irrelevant theory, questionable conclusions, and has kept statisticians from working on a large range of interesting current problems. Algorithmic modeling, both in theory and practice, has developed rapidly in fields outside statistics. It can be used both on large complex data sets and as a more accurate and informative alternative to data modeling on smaller data sets. If our goal as a field is to use data to solve problems, then we need to move away from exclusive dependence on data models and adopt a more diverse set of tools.

(Breiman, 2001)

The 21st century has brought continued exponential growth in technologies for collecting, storing, processing, and analyzing data. We already have noted Laney's (2001) identification of the three Vs to describe the challenges of the data deluge. It was left to Chris Anderson, then editor-in-chief of *Wired*, to bring data science into mainstream thinking about the opportunities and challenges inherent in the rapid accumulation of data of all kinds. In his provocative 2008 editorial, "The end of theory: The data deluge makes the scientific method obsolete," he argued that data science, by virtue of its reliance on computers, algorithms, and big data, was capable of generating more useful and accurate insights than domain experts using traditional methods to develop and test carefully constructed hypotheses (Anderson, 2008). Anderson's challenge to traditional market, opinion, and social research reverberates through discussions of big data to this day.

Over about the last decade, the legitimacy of data science as a field of study and as a profession has been firmly established. There are now journals, professional associations, and university programs all focused on data science or as it is sometimes simply called, "analytics." Writing on the Dataists blog, Mason and Wiggins suggest a loose but useful definition of data science, as currently practiced. It echoes the main themes described by Wu two decades earlier.

Data science is ... a blend of the hackers' arts; statistics and machine learning; and the expertise in mathematics and the domain of the data for the analysis to be interpretable ... It requires creative decisions and open-mindedness in a scientific context.

(Mason and Wiggins, 2010)

One common critique of the data scientist's reliance on automated analytic methods is that it can lead to false conclusions and incorrect models. This is expressed in multiple ways—spurious correlations, endogeneity, causation versus causality, and so forth (see, e.g., Fan and Liu, 2014; Janju, 2013; Wright, 1921). The key point is that it is not enough to simply be good at statistics. Domain expertise also is critical as is simple good judgment so that we do not blindly accept whatever the algorithms tell us. To quote Nate Silver, "Data-driven predictions can succeed—and they can fail. It is when we deny our role in the process that the odds of failure rise" (Silver, 2012).

Closer to home, Shearer (2007) placed data science into the context of market, opinion, and social research by describing these four characteristics:

- 1) Use of a data-driven approach in place of a user-driven approach. Rather than the researcher deciding which variables to analyze, algorithms explore the data and with little guidance from the researcher.
- 2) In place of a limited number of variables and relationships, data mining makes it possible to include a very large number of variables among which there may be complex and multiple relationships.

- 3) Data miners typically do not focus on hypothesis testing, rather they “let the data tell the story,” meaning that the analysis is done by algorithms without regard to assumptions about the relationship among data items.
- 4) Survey researchers typically have reported their results in tables, reports, and graphs; data miners are more focused on building predictive models.

This short history might convey the impression that the evolution of techniques for collecting, storing, and analyzing very large datasets was primarily within the context of market research and business intelligence. That has not always been the case. There are multiple examples in the U.S. public sector as well, especially in the post 9/11 era when it has been extremely controversial on grounds of privacy invasion and government intrusion (Mohammed and Goo, 2006). Big data and data science opportunities exist in the physical sciences as well and have been widely used in fields as diverse as cosmology and weather forecasting (Reed, 1999).

3.4 Assessing Data Quality

Gurwitz (2012) argues that for big data to be useful to market, opinion, and social researchers, it must meet four criteria:

- 1) It must be valid, that is, “as accurate as possible an indicator of the phenomenon being measured.”
- 2) It must be complete with as little missing data as possible. Where there are missing data, the missingness should be random rather than systematic.
- 3) It must be representative of the population of interest, however defined.
- 4) It should be at the lowest possible unit of analysis, preferably at the individual level.

Three of these criteria—validity, missingness, and representativeness—are familiar to survey researchers and resemble the general themes within the total survey error framework (Groves, 1989). We are accustomed to applying them in evaluations of survey data quality. But survey researchers typically have a significant advantage in that the most important survey datasets tend to be well documented, sometimes painfully so. Most of the information needed to assess bias and make data quality judgments is known. Such is not always the case with big data.

3.4.1 Validity

As previously noted, much of the value of big data is realized through combining from multiple sources or the repurposing of data collected or generated for some purpose other than research. As described earlier in this chapter, the original sources may have errors. In addition, by the time those data reach the researcher, they likely will have undergone multiple stages of processing, such as data reductions, edits, transformations, merges, and additional reductions. Tracing the path of a single data element from its original collection to the analytical dataset can be difficult and, in some cases, impossible.

In an ideal world the data we use will have been fully curated throughout its lifecycle and its provenance known. Even curated data may be simply an input to additional processes downstream aimed at further enhancing value through merging in of other data of different provenance. As just described earlier, the heart of this process is a series of often very complex ETLs, each with the potential to introduce errors and accumulate them with additional processing, undermining validity. Working one’s way through all of this to develop at least some level of confidence that a data item measures what it purports to measure can be extremely difficult

and in some cases be impossible. As powerful as big data may seem in the abstract, in practice it can be very messy.

3.4.2 Missingness

Merging of disparate databases each contributing some additional bits of information invariably creates “holes” in individual data records. For example, data brokers typically acquire data on individual households from a wide array of sources. They use these data to design products that are described as having dozens of different items (variables) on a very high proportion of U.S. households. Yet some of these items often have significant levels of missing data. The more items included, the greater the likelihood of having more missing data. This same phenomenon occurs in any research project that attempts to leverage a variety of sources on the same entity, be it an individual, a household, a company, or an institution. There are a variety of fusion and imputation techniques and analysis options to help us deal with this, but all run the risk of reducing accuracy in the eventual estimates.

3.4.3 Representation

One of the great myths of big data is its characterization of “N = All” (Janju, 2013; Mayer-Schonberger and Cukier, 2013). Closely allied with this thinking is the misbelief that even though a dataset may not cover a large enough proportion of the population of interest, having such a large dataset washes out any bias due to limits in coverage or measurement error. Granted, there are datasets with outstanding coverage of populations such as those receiving some benefit (e.g., Social Security or Medicare in the U.S.A.) or everyone who took out a mortgage at a specific bank. Customer files may have poorer coverage depending on the product sold. POS data have coverage limits because not all stores use POS systems.

Other sources for big data analytics may have even poorer coverage. Consider social media data or much of the data currently being generated by the IoT. These kinds of data sources, especially social media, can provide a level of depth and granularity to larger databases with better coverage. But, as Gurwitz points out, when multiple datasets are merged, the effective sampling frame typically is the smallest of the sourced datasets (Gurwitz, 2012). Thus, the more data we introduce, the less representative it can become.

The process of assessing the quality of big data is far more complex than with survey data. This is the tension described by Groves (2011) when comparing designed (surveys) versus organic data (mostly administrative data). The refined process of designing, conducting, and analyzing surveys offers a level of control (and perhaps comfort) that we lose with big data. A new model for assessing quality is required. See Chapter 1 for a discussion of this concept, referred to there as total survey error.

3.5 Applications in Market, Opinion, and Social Research

Looking back over time, researchers often have used customer data in commercial companies and administrative data in national statistical organizations as sampling frames. The more interesting relationship between large databases and surveys, regardless of sector, is the combination of the two to create richer datasets and ultimately greater insight and more value.

3.5.1 Adding Value through Linkage

The most successful data mining and customer relationship management (CRM) projects of the 1990s were those that were able to combine data from different data silos throughout an organization. Especially in very large companies, there was an historical tendency for individual departments to amass data around their view of the customer or market. In an auto company, for example, there might be one database that tracked warranty claims or service visits, another for financial transactions such as loan payments, and still another tracking sales to individuals or performance of dealers. Sometimes these databases might contain unique data and other times duplicative data. Combining them not only improved efficiency and accuracy, but it also offered the potential for an enhanced view of the customer that, for example, might make sales promotions more effective or identify those customers most likely to defect to another brand. Despite the potential advantages, conflicting database formats, definitions of data items, and simple parochialism often made this difficult.

Smarter use of more powerful technologies now makes it possible to combine different databases not just within a company but also across companies. Data brokers such as Acxion, Equifax, and TransUnion collect a wide range of information on individual consumers (e.g., credit card purchases, magazine and newspaper subscriptions, financial histories, and demographics) that can be matched in and merged with a company's customer records to not just understand their customers better but also take specific actions such as targeting offerings or denying credit. As consumers and governments become more concerned about privacy and the commercial use of personal data, companies like these are under increased scrutiny (Markey, 2015). Still, as of this writing, they are an extremely important part of the big data ecosystem.

There also are less ominous examples of companies cooperatively sharing big data to their mutual advantage. For example, the Weather Company collects about 20 terabytes of weather data every day for weather stations all over the world (Henschen, 2013). Mostly the company uses those data to develop and feed the models that make up the company's core business of forecasting the weather for virtually every location on the planet. More recently it has begun to use those same data to support real-time marketing of products ranging from air conditioners to bug spray (Rossman, 2013). It began by persuading retail and consumer goods manufacturers such as Walmart and Procter & Gamble to give them sales data for every product they sold, in every store, in every aisle, all over the U.S.A. Using the massive processing power that big data applications have made famous, the company is able to find correlations between weather and sales of tens of thousands of products and brands, turning out predictions of which products are likely to sell best given the expected weather in geographic areas down to the zip code level (Suddath, 2014).

Similar opportunities exist within the public sector and especially national statistical organizations (Strujis et al., 2014). One of the best known is the use of administrative data in the Nordic countries to produce census-type statistics without the use of standard census methods, that is, interviewing people (Wallgren and Wallgren, 2014). There are long-standing practices in Norway, Sweden, Finland, and Denmark of compiling and maintaining individual level data about people and businesses on a wide range of topics that includes demographic characteristics, property ownership, educational achievement, morbidity and mortality, building assessments, wages paid, number of employees, taxes paid, and so on. Linking these data makes it possible to generate a wide range of national statistics without the need for census-type data collections. Similar opportunities may exist in other countries including Switzerland, Italy, and Austria. Still other countries are actively considering transforming their traditional census to one relying primarily if not solely on administrative records (Ralphs and Tutton, 2011).

In the U.S.A., the practice of linking administrative records to enhance the value of data goes back over 40 years. As its name implies, the 1973 CPS-IRS-SSA Exact Match Study was an undertaking of three U.S. government agencies (the Bureau of the Census, the Internal Revenue Service, and the Social Security Administration (SSA)) aimed at matching the records of sampled individuals from the 1973 March Supplement of the Current Population Survey to their social security earnings records and selected items from the 1972 Federal income tax returns (Kilss and Scheuren, 1988).

Each party had its own interests in the project. The SSA was primarily interested in augmenting its data to provide a more accurate basis for analysis of the impact of potential policy changes on income distribution. In subsequent years and up to the present day, SSA designs and conducts similar studies, although those studies rely primarily on statistical matching (also known as data fusion) rather than the exact matching of personally identifiable information used in 1973. (Radner and Daniel, 1988).

The U.S. Census Bureau has a long-standing interest in using the data from other agencies to improve its survey design and implementation processes. Matching studies were used in censuses going back to 1950. That tradition has continued since and is now routinely used in a number of census-conducted surveys in addition to the decennial. See Chapter 21 for a discussion of uses of big data by the U.S. Census Bureau and other national statistical institutes.

3.5.2 Combining Big Data and Surveys in Market Research

ESOMAR, the worldwide association for market, opinion, and social research, defines market research in somewhat lofty terms as “the systematic gathering and interpretation of information about individuals or organisations using the statistical and analytical methods and techniques of the applied sciences to gain insight or support decision making” (ESOMAR, 2008). Those of us who labor daily in the market research vineyard see our work in less prosaic terms, namely, helping clients improve their products and services so that they can sell more of them, either through tactical or strategic advice.

Market researchers historically have used a range of quantitative (mostly surveys) and qualitative techniques (mostly focus groups) in their work. The advent of new sources of customer opinion and behavior (first social media and then big data) has been viewed both as a threat and an opportunity. Some worry that big data eventually will obviate the need for surveys, much as the use of registry systems in Nordic countries has made traditional censuses less necessary. Others see big data and surveys as complementary (Japac et al., 2015). In one view, big data increases the amount of data available, contributing information to our analyses that increases their validity. Another view characterizes big data as describing what people do while surveys help us understand why they do it.

Consider two examples. The first, described by Gurwitz (2012), is the relatively common problem of attempting to predict likely purchase behavior using a sample of customers and based on a standard set of attitudinal questions around product preferences, price, selection, and so on. Gurwitz showed that a relatively simple regression model based on attitudinal measures was improved substantially by the addition of a modest amount of behavioral data from the client’s customer files and the use of more robust regression methods, resulting in an increase in the R^2 from 0.42 to 0.85.

Mishkin (2014) described a more complex case involving what he calls a “continuous improvement cycle” based on the use of traditional quantitative and qualitative methods in conjunction with data from his client’s customer files. In the demonstration case the client was a mobile service provider, and the primary goal of the research was to develop a strategy to reduce churn,

that is, customer defections to another provider. By identifying those customers who were most at risk, his client was able to take defensive actions aimed at retaining them and reducing churn.

The process Mishkin describes is not unlike that used in political campaigns under the general heading of “microtargeting” (Korman, 2001). This technique uses publicly available files of registered voters and the massive databases of commercial data brokers to develop segmentation schemes that classify voters into groups (“segments”) based on combinations of demographic, attitudinal, and behavioral characteristics. It starts with a survey of a sample of registered voters who are interviewed to assess their political attitudes and voting intentions, also collecting a variety of lifestyle behaviors and demographics. These data are used to build a segmentation model that can be matched using demographics and lifestyle indicators to data broker files containing information on almost all U.S. households. Having then segmented most of the U.S. population, the campaign can design its voter outreach to appeal to each segment. It is not an exaggeration to say that every voter is assigned a score that allows the campaign to not just estimate his or her voting propensity but also information about what motivates that individual’s preferences for one candidate over another.

3.6 The Ethics of Research Using Big Data

Two of the most important pillars of the ethical foundation of market, opinion, and social research are consent and confidentiality. Consent means that prospective research participants are provided with a description of the purpose of the research and how their data will be used. Confidentiality means that the researcher will provide the level of data protection needed to ensure that the identity of an individual participant is never disclosed to any party without the explicit consent of that participant and that any data released for analysis will be anonymized.

The evolving world of big data is problematic on both fronts. When we reuse data collected for some other purpose we have an obligation to determine whether our planned use of the data is consistent with the terms under which individuals agreed to provide them. Further, recent research has demonstrated that the amount of data now available on individuals and the processing power to combine and analyze it has rendered traditional approaches to anonymizing obsolete (see, e.g., de Montjoye et al., 2014; Sweeney, 2013).

Both of these topics are beyond the scope of this chapter. They are raised here because they are an important element of the big data debate. For further discussion see Japiec et al. (2015), Lane et al. (2014), and Chapter 4 of this volume.

3.7 The Future of Surveys in a Data-Rich Environment

Market researchers, in particular, spend a great deal of time these days thinking about and forecasting the future of the broader research industry (see, e.g., Kaden et al., 2012). There is some consensus forming around three transitions that over time will transform what we do.

First, it is widely believed that we are moving from a world where data are scarce to one in which they are plentiful. Put simply, this is the difference between surveys and the world created by the embedding of technology in all aspects of our daily lives, a world in which entire research reports can be compiled through a series of Google searches or an emerging set of “small data” applications (Bonde, 2013a, b) that provide online access to very large databases (see, e.g.,

McIntyre, 2012; van Panhuis et al., 2013). From this perspective, the need for surveys will decline because the data already will exist, waiting to be retrieved and analyzed.

Second, research will focus less on gathering data by asking questions and more on observation and listening. The increasing and widespread acceptance of principles of cognitive psychology such as dual process theory (Kahneman, 2011) has led many to argue that behavior is a more reliable predictor of the choices people make than asking about attitudes and intentions as is typically done in surveys. It also reflects a view of social media as an unfiltered source for understanding what is important to people and the questions we should be asking. All components of big data, including the IoT, are a potential source of insight about what people believe and how they behave.

Third, as the data sources available for studying a topic multiply, researchers will emphasize synthesis of data from multiple sources and methods over analysis of a single dataset. Smith and Fletcher characterize this difference as "... drawing together different types of desk, observation, qualitative, and quantitative evidence into a single combined whole, rather than just analyzing solitary pieces of uncoordinated evidence" (Smith and Fletcher, 2004). Data points from different sources that conflict despite purporting to measure the same thing will need to be examined and reconciled. Decisions both in the private and public sectors will rely on multiple data points with no one source viewed as consistently reliable.

In this world, surveys become just one of many ways to do research. When Joan Lewis, Consumer and Market Knowledge Officer at Procter & Gamble, was asked whether social media would replace surveys, she responded, "We need to be methodology agnostic" (Neff, 2011). By that she meant that there are multiple methods and data sources that might be used to study a problem.

We now are seeing an expanding toolkit of methods and data sources from which a researcher can choose those that best fit the phenomenon he or she is investigating. These include but are not limited to social media analytics, online communities, biometrics, prediction markets, facial coding, virtual reality, crowdsourcing, and a variety of mobile-based techniques, both quantitative and qualitative (Greenbook, 2015). The job of the researcher is to select the best match to satisfy the requirements—the topic, the required level of precision, the available budget, and the immediacy of the need.

Given this outlook, there is no shortage of people predicting that a key consequence of these changes is the death of surveys (see, e.g., O'Connor, 2010; Poynter, 2014). There is also evidence that surveys are already slowly losing their historical dominance. ESOMAR estimated that in 2008 the combination of surveys by telephone, postal, face-to-face, and online comprised 55% of the total spend on market research worldwide (ESOMAR, 2009). By 2014 that number had fallen to 45% (ESOMAR, 2015). They point to new players specializing in predictive analytics and management consulting that are encroaching on the turf formerly occupied by traditional research companies and organizations. We also have discussed in this chapter the increasing use of administrative data in place of survey methods by national statistical organizations.

Science has a key role to play as our profession works its way through all of this. The first question that needs to be asked of any methodology is whether it measures what it purports to measure. Is there some theoretical basis or at least a fully explicated model that can be examined and validated? And, even if that foundation exists, how do we judge the validity of a set of results in a specific use of that method?

For this we need a framework that specifies the potential sources of error that might compromise our results. The total survey error model has been that framework for surveys. A further elaboration into a much broader set of considerations is essential as we move to a more diverse set of methods. Its application beyond traditional surveys is a key goal of this book.

The marketplace—broadly defined as users of data across all sectors of the economy including academia and government—also plays an important role. It ultimately falls to the data user to decide whether a specific method or set of data is the right fit to the decision that needs to be made or the phenomenon being examined. Biemer and Lyberg describe this in the survey context as “fitness for use” with the three dimensions of accuracy, timeliness, and accessibility. Science informs this decision, but it does not dictate it (Biemer and Lyberg, 2003).

The long-term viability of surveys may well rest on their designed character, their ability to target a specific population, to specify the data of interest, and to design a collection process that minimizes error. As we have seen, there is an element of taking what you can get with found data, and issues of coverage, meaning, and accuracy are a constant concern. The future of big data, and perhaps that of the survey profession, rests on how well we solve these problems.

For now, the most prevalent view seems to be one that stresses the complementarity of big data and surveys (Forsyth and Boucher, 2015; Macer, 2015). Not surprisingly, this viewpoint is forcefully made in the “AAPOR Report on Big Data” (Japac et al., 2015) but it also is shared among market researchers where enthusiasm for big data is strongest. In 2015, ESOMAR interviewed the CEOs of the four largest global market research companies (Nielsen, Kantar, Ipsos, and GfK) to get their views on the future of research in a data-rich world. All agreed that big data by itself is not enough. To quote Didier Truchot, CEO of Ipsos, “I’m totally convinced that in 5 or 10 years from now we will still need a lot of grounded research through surveys” (Bowman, 2015).

Groves (2016), on the other hand, sees surveys as not just useful but essential, especially in the realm of official statistics. He argues that many big data sources have two problems. The first is that they often have few variables and extracting their value requires linkage to other data. Think of data from the IoT or POS scanner data. The second problem is lack of representativity. Think of social media or one of any number of big data sources lacking full coverage of the population of interest. In the first case, the survey designer’s ability to control content can provide much-needed context for big data sources where context is missing. In the second, surveys with their reliance on samples capable of targeting specific populations can be used to improve representativity.

As comforting as that may be for readers of this book, it does not free us from the responsibility to broaden our understanding of data sources and methods beyond surveys. To draw on Couper’s analogy (2013), surveys are like screwdrivers with different uses and capabilities. But they are not the only tools in our kit. There are also hammers, pliers, wrenches, and drills with each designed for a task that screwdrivers cannot do. To do a job right, we use them in combination. The challenge to our profession is to learn when and how to use the right tool(s) for the job at hand. That may be a tall order for those of us accustomed to a survey as the always-obvious choice.

The more difficult challenge posed by big data may not be coming to grips with all those Vs. Rather, it may be in accepting the validity of the analytic approaches of the data scientist. To once again quote Chris Anderson:

At the petabyte scale, information is not a matter of simple three- and four-dimensional taxonomy and order but of dimensionally agnostic statistics. It calls for an entirely different approach, one that requires us to lose the tether of data as something that can be visualized in its totality. It forces us to view data mathematically first and establish a context for it later.

(Anderson, 2008)

The discipline of survey research historically has relied on a set of standard approaches to data analysis as well as collection. Those analytic approaches can be used with big data, but in doing

so we risk losing much of what it has to offer. As Tukey pointed out a half century ago, the use of statistics to infer population characteristics from samples is just one technique for analyzing data. There are others. To take advantage of them, "we must give up the idea that ... if data analysis cannot be a logico-deductive system, it inevitably falls to the state of a crass technology" (Tukey, 1962).

Big data has the potential to revolutionize market, opinion, and social research by giving us both more data and better tools. The question is not *if* it will happen but *how long* it will take. Whether there is a long-term role for surveys and what that role might be are more difficult questions. The important thing is that we not let them get in the way of moving ahead.

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